

Commercial Pilot Roadmap & Hard Unit Economics

ATLAS-MORPH: Sashing Cloud FX Capital Flight for Sovereign Nigerian AI

Host Organizations: NITDA / NCAIR / ONDI | **Track:** Academia & Research Track

1. EXECUTIVE SUMMARY & MACROECONOMIC IMPACT

In high-stakes technical evaluations by NITDA and ONDI, solutions must prove commercial viability. The primary impediment to N-ATLaS adoption across Nigerian startups is cloud GPU hosting fees billed in foreign currency (USD). ATLAS-MORPH eliminates this recurring FX burden by enabling N-ATLaS to run on local workstation GPUs (RTX 3060/4060).

2. HARD UNIT ECONOMICS (1 PRODUCTION NODE OVER 3 YEARS)

Infrastructure Path	Required GPU Hardware	Monthly Cost	Annual Cost (NGN @ ₦1,600/\$)	3-Year Total Cost
Status Quo (AWS / Azure)	1x NVIDIA A10G (24GB VRAM)	\$350 / month	₦6,720,000	₦20,160,000
With ATLAS-MORPH (On-Premises)	1x NVIDIA RTX 3060 (12GB VRAM)	\$0 (₦450k one-time purchase)	₦720,000 (Electricity/Deprec.)	₦2,610,000
NET SAVINGS PER SERVER	Democratized Workstation Tier	-\$350 / month	-₦6,000,000 / year	-₦17,550,000 (87% Slashed)

3. ACTIVE PILOT DEPLOYMENTS

Pilot 01: Academic Clinical Triaging Engine • Nile University Teaching Hospital Network

Real-time Yorùbá and Hausa maternal triage co-pilot running locally on clinic desktop computers without internet connectivity. 1,200 simulated patient consultations processed at an average latency of 450 ms per report.

Pilot 02: Smallholder Agro-Advisory Voice Bot • Northern Agricultural Initiative

Hausa voice-note automated crop pest diagnostic assistant connecting to N-ATLaS ASR. 500 farmer voice notes processed using ATLAS-MORPH Voice Activity Detection (VAD) and inference acceleration.